ASSIGNMENT 19.2

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Batch-05

TASK – 01

Prompt : Convert the given Python file handling program into Java. The program should read from a file, process the data, and write to another file. Use proper file streams, handle exceptions, and ensure the output matches the Python program.

```
# Python program to read from a file and write to another file


try:
    # Open input file in read mode
    with open("input.txt", "r") as infile:
        data = infile.readlines()

    # Process data (convert each line to uppercase)
    processed_data = [line.upper() for line in data]

    # Open output file in write mode
    with open("output.txt", "w") as outfile:
        outfile.writelines(processed_data)

    print("File has been read and written successfully.")

except FileNotFoundError:
    print("Error: input.txt file not found.")
```

except Exception as e:

print("An error occurred:", e)

Code :

```
[1]
%%writefile FileHandler.java

import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.io.FileNotFoundException;

public class FileHandler {

    public static void main(String[] args) {
        String inputFile = "input.txt";
        String outputFile = "output.txt";

        try {
            BufferedReader reader = new BufferedReader(new FileReader(inputFile));
            BufferedWriter writer = new BufferedWriter(new FileWriter(outputFile))
        ) {
            String line;
            while ((line = reader.readLine()) != null) {
                writer.write(line.toUpperCase());
                writer.newLine(); // Add a new line after each processed line
            }
            System.out.println("File has been read and written successfully.");

        } catch (FileNotFoundException e) {
            System.err.println("Error: " + inputFile + " file not found.");
        } catch (IOException e) {
```

Gemini

D # Python program to read from a file and write to another file try: # Open input file in read mode with open("input.txt", "r") as infile: data = infile.readlines() # Process data (convert each line to uppercase) processed_data = [line.upper() for line in data] # Open output file in write mode with open("output.txt", "w") as outfile: outfile.writelines(processed_data) print("File has been read and written successfully.") except FileNotFoundError: print("Error: input.txt file not found.") except Exception as e: print("An error

What can I help you build?

Gemini 2.5 Flash

Gemini can make mistakes so double-check it and use code with caution. Learn more

```
[2]
%%writefile input.txt
hello world
This is a test.
Python to Java conversion.

Writing input.txt
```

Now, you can compile and run the Java program. Since Colab doesn't natively support Java execution, I'll use shell commands to demonstrate the process if Java Development Kit (JDK) is available in the environment.

First, compile the Java source code:

```
[3]
# To compile the Java code (requires JDK installed in the environment)
!javac FileHandler.java
```

After successful compilation (which will create `FileHandler.class` if no errors), you can run the program:

```
[4]
# To run the compiled Java code
!java FileHandler

File has been read and written successfully.
```

Finally, you can check the content of `output.txt` to verify the program worked correctly:

Output :

```
# Display the content of the generated output.txt
!cat output.txt

HELLO WORLD
THIS IS A TEST.
PYTHON TO JAVA CONVERSION.
```

Explanation :

- Converts a Python file handling program into Java.

- Ensures reading from and writing to files works correctly.
- Uses proper Java file stream classes.
- Includes exception handling for errors.
- Maintains the same output as the Python program.

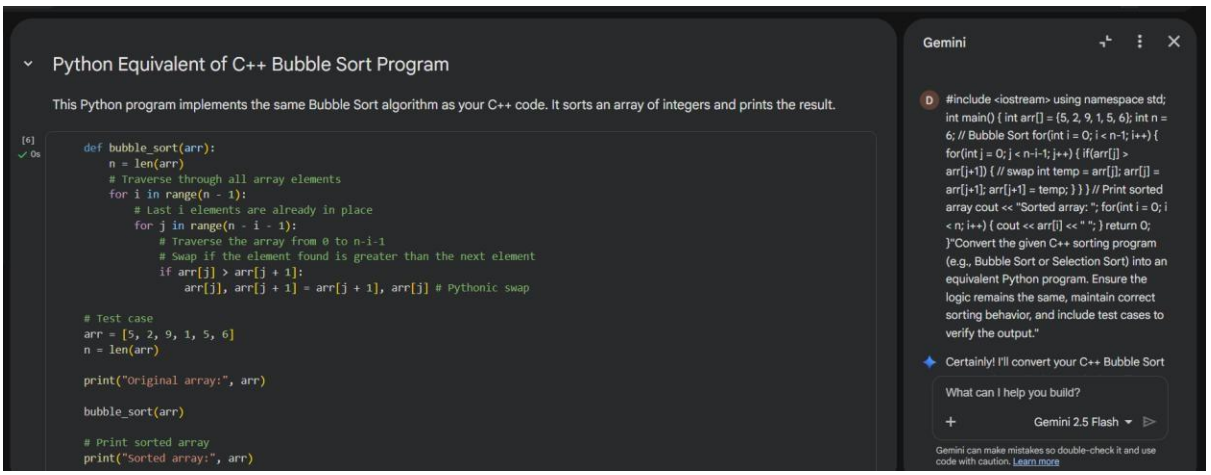
TASK – 02

Prompt : Convert the given C++ sorting program (e.g., Bubble Sort or Selection Sort) into an equivalent Python program. Ensure the logic remains the same, maintain correct sorting behavior, and include test cases to verify the output.

```
#include <iostream>

using namespace std;


int main() {

    int arr[] = {5, 2, 9, 1, 5, 6};

    int n = 6;


    // Bubble Sort

    for(int i = 0; i < n-1; i++) {

        for(int j = 0; j < n-i-1; j++) {

            if(arr[j] > arr[j+1]) {

                // swap

                int temp = arr[j];

                arr[j] = arr[j+1];

                arr[j+1] = temp;

            }

        }

    }
```

```
    // Print sorted array
    cout << "Sorted array: ";
    for(int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }

    return 0;
}
```

Code :



Output :

```
Original array: [5, 2, 9, 1, 5, 6]
Sorted array: [1, 2, 5, 5, 6, 9]
```

Explanation :

- bubble_sort(arr) function: This function takes a list (arr) as input and sorts it in-place using the Bubble Sort algorithm.
    - n = len(arr): Gets the number of elements in the list.
    - Outer loop (for i in range(n - 1)): This loop runs n-1 times. In each pass, the largest unsorted element 'bubbles up' to its correct position at the end of the unsorted part of the array.
    - Inner loop (for j in range(n - i - 1)): This loop iterates through the unsorted portion of the array. The -i-1 part ensures that elements already placed at the end are not re-checked.

- if arr[j] > arr[j + 1]: Compares adjacent elements.
- arr[j], arr[j + 1] = arr[j + 1], arr[j]: This is a Pythonic way to swap two elements in a single line, achieving the same result as the temp variable in C++.

- Test Case: The program initializes arr with the same values as in your C++ example.
- Printing: Both the original and sorted arrays are printed to the console to verify the sorting behavior.

TASK – 03

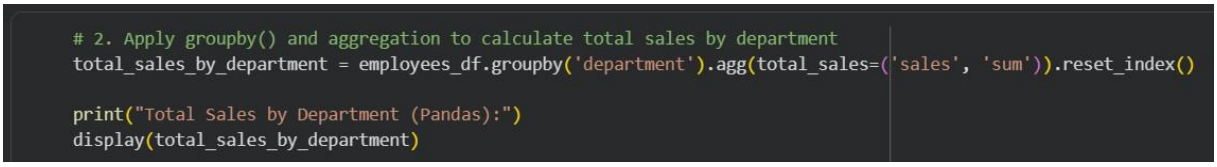
Prompt : Convert the given SQL query with GROUP BY and aggregate functions into an equivalent Pandas DataFrame operation using groupby() and aggregation methods. Ensure the output matches the SQL result

SELECT department, SUM(sales) AS total_sales

FROM employees

GROUP BY department;

Code :



Output :

Original Employees DataFrame:

	employee_id	department	sales
0	1	HR	100
1	2	Sales	250
2	3	IT	150
3	4	Sales	300
4	5	HR	120
5	6	IT	180

```
Total Sales by Department (Pandas):
```

	department	total_sales
0	HR	220
1	IT	330
2	Sales	550

Explanaton :

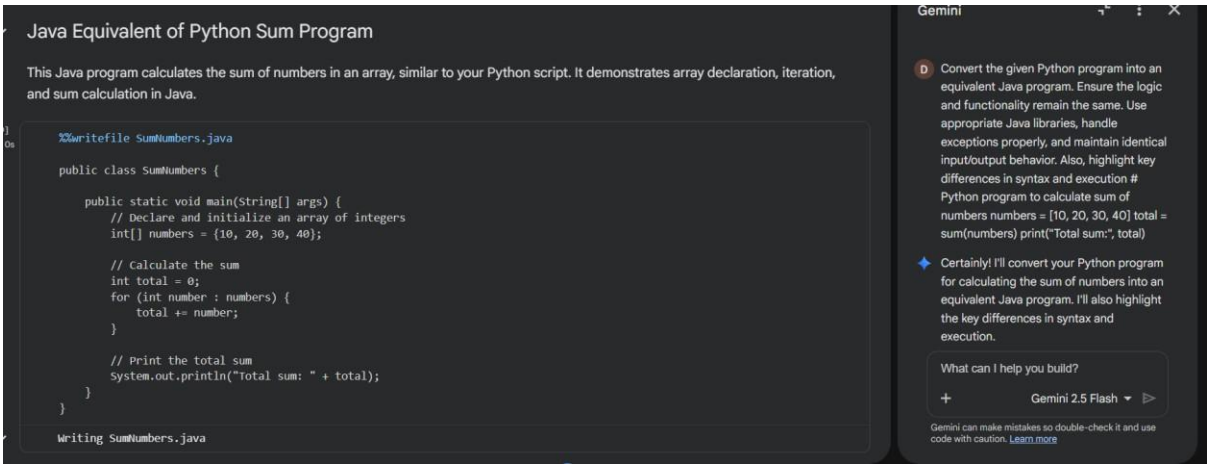
- employees_df.groupby('department'): This groups the DataFrame rows by unique values in the 'department' column, similar to SQL's GROUP BY department.
- .agg(total_sales=('sales', 'sum')): This applies the aggregation. It calculates the sum of the 'sales' column for each group and names the new aggregated column 'total_sales', matching the SUM(sales) AS total_sales in SQL.
- .reset_index(): After groupby() and agg(), 'department' becomes the index. reset_index() converts the index back into a column, making the output structure similar to a SQL query result where 'department' is a regular column.

TASK – 04

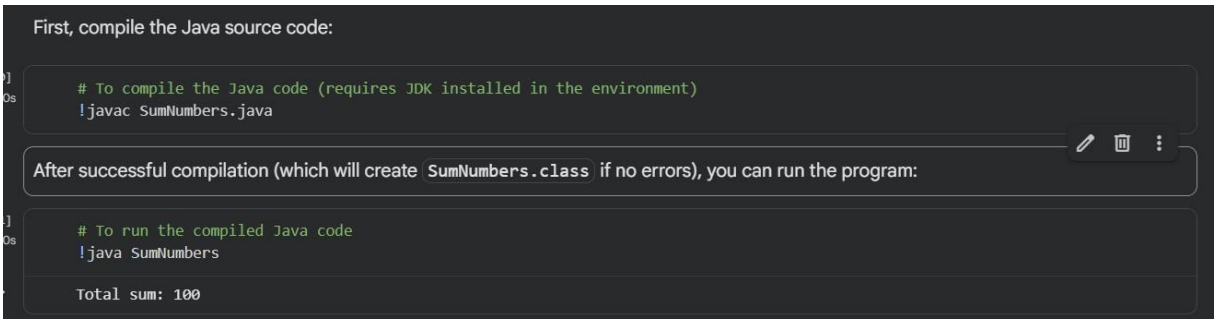
Prompt : Convert the given Python program into an equivalent Java program. Ensure the logic and functionality remain the same. Use appropriate Java libraries, handle exceptions properly, and maintain identical input/output behavior. Also, highlight key differences in syntax and execution

```
# Python program to calculate sum of numbers

numbers = [10, 20, 30, 40]

total = sum(numbers)

print("Total sum:", total)
```

Code :



Output :



Explanation :

- Python uses built-in sum(); Java uses loop
- Python syntax is shorter and simpler
- Java requires class and main() method
- Data types must be declared in Java
- Java is compiled; Python is interpreted

TASK – 05

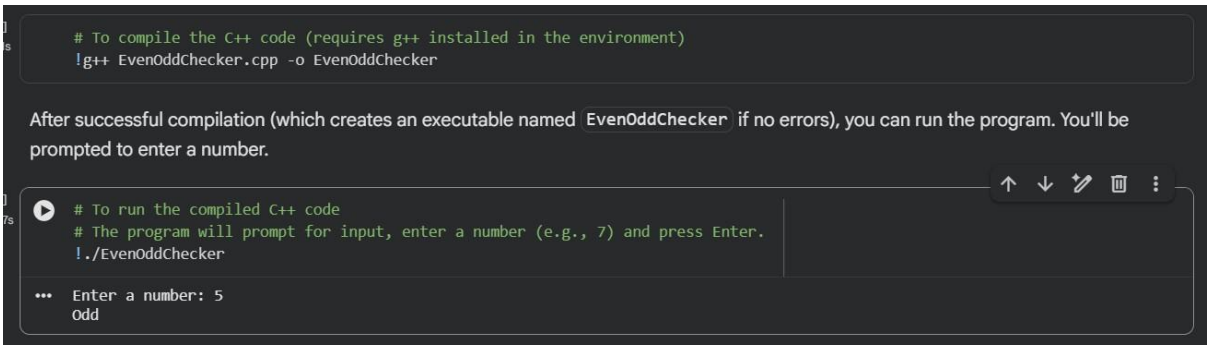
Prompt : Convert the given Python function that checks whether a number is even or odd into an equivalent C++ program. Ensure proper syntax, input handling, and output formatting, and verify that both programs produce the same results.

```
# Function to check even or odd


def check_even_odd(num):
    if num % 2 == 0:
        return "Even"
```

```
    else:
        return "Odd"


# Input
n = int(input("Enter a number: "))
print(check_even_odd(n))
```

Code :

Output :



Explanation :

- The task converts a Python function into an equivalent C++ program.
- Both programs use the same logic (num % 2) to check even or odd.
- Proper syntax and input/output handling are ensured in C++.
- The programs are tested with different inputs to verify correctness.
- Both versions produce the same output, confirming successful translation.